1. Clinical Question & Core Hypothesis

Clinical Question

Why do some high-risk localized prostate cancers with abundant lymphocyte infiltration nonetheless fail to respond to intensified systemic therapies? The Ontologist synthesis highlights a triad—tumor-intrinsic genomic risk (Decipher GC), the functional immune state (IO predictive signatures), and systemic inflammation/metabolic context (EDIH/insulin signals)—that could explain this paradox?

Working Hypothesis

Among men with high-risk localized prostate cancer, the predictive value of tumor-infiltrating lymphocytes (TILs) for benefit from intensified systemic therapy is context-dependent. We hypothesize that: (1) Decipher high-risk tumors with abundant TILs and an exhausted immune phenotype (IO exhaustion signature) will exhibit reduced response to intensified neoadjuvant/systemic therapy due to T cell dysfunction and supportive myeloid/immunometabolic remodeling; (2) Decipher low-risk tumors with high TILs and an active effector IO signature will show greater therapeutic benefit from intensified regimens; (3) Secondary modifiers (CSF-1–driven macrophage remodeling, PD-L1 checkpoint signaling, PTGER4 signaling, CENPA-associated proliferative drive, EDIH/metabolic/inflammatory signaling, and insulin_response_markers) will modulate the magnitude of the TIL–Decipher–IO interaction by either dampening or enhancing T cell function and therapy responsiveness. Accordingly, the primary endpoint (therapy_response) will be driven by a three-way interaction: TIL density × Decipher GC status × IO exhaustion state, with secondary modifiers shaping the strength and direction of this interaction. This framework predicts: a) TIL-high/Decipher-high with IO exhaustion will have the lowest therapy_response rates; b) TIL-high/Decipher-low with IO effector signatures will have the highest response rates; and c) Tier-2 modifiers will attenuate or amplify these effects in a dose-dependent manner.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	Decipher GC, IO predictive signatures
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Decipher GC: Decipher GC path (genomic risk stratification) and the IO predictive signatures path (immune functional state) because together they provide an interpretable tumor-intrinsic vs.

IO predictive signatures: IO predictive signatures), and systemic inflammation/metabolic context (EDIH/insulin signals)—that could explain this paradox.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CSF1	Supporting modifier from knowledge graph
Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	PTGER4	Supporting modifier from knowledge graph
Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	EDIH	Supporting modifier from knowledge graph
Secondary	insulin_response_markers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CSF1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.